

CHAPTER XIX

CLASSICAL OPTIMIZATION TECHNIQUES

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1. Introduction

Optimization simply means to find the best possible solutions out of the available alternatives under the given circumstances. Optimization Techniques are very important in solving real life problems and various optimization techniques have been successfully used in different fields to solve problems. Every optimization problem has three main components: an objective function, independent variables, known as decision or design variables, and constraints. A physical problem can be mathematically expressed as an objective function of either one or more than one independent variables. The classical techniques are used for finding the optimum solution for optimization problems. According to the classical optimization theory these techniques operate under the assumption that the derivatives are continuous and the function is twice differentiable. These analytical techniques employ differential calculus procedures to locate the optimum positions. The practical uses of classical optimization techniques are restricted because certain practical subjects incorporate non-continuous and perhaps non-differentiable functions. The development of calculus-based optimization techniques provides a base for most numerical optimization techniques and helps to overcome the limitations of classical optimization methods (Cesari, 2012). Typically, numerical optimization

techniques begin from an arbitrary starting point that is chosen within a certain search space, near the real solution, and proceed iteratively toward the optimum solution. Optimization issues can be categorized according to the type of objective function that is in operation. When the objective function is nonlinear, gradient-based numerical techniques and elimination methods are recommended to be used. Regarding the gradient-based techniques, numerical optimization algorithms have two main parameters to select: the length of each step and the directional path of the search. On the other hand, elimination techniques can also be used to figure out the optimal value of the complicated unimodel that involves non-continuous and possibly non-differentiable functions (Nocedal & Wright, 1999).

This chapter provides the required conditions for necessity and sufficiency for a local maximum or local minimum of both single and multivariable classical optimization techniques, along with examples to find the optimal solution. Furthermore, gradient-based numerical optimization techniques, including Steepest Descent and Newton, as well as elimination techniques, including Fibonacci in addition to Golden Section methods, are discussed.

2. The Required Conditions for Necessity and Sufficiency of Classical Optimization Techniques

The required conditions for necessity and sufficiency for the local optimum of a single-variable function are given by the next two theorems (Reklaitis, et al., 1983):

Theorem 2.1. If a function of one variable $f(x)$ is defined in the interval $a \leq x \leq b$ and has a local minimum at $x = x^*$, where $a < x^* < b$, and if the derivative $\frac{df(x)}{dx} = f'(x)$ exists as a finite number at $x = x^*$, then $f'(x) = 0$, it's necessary, which is still not adequately sufficient, in order to determine the function's optimum positions.

Theorem 2.2. Let the derivatives of a single-variable function

$$f'(x^*) = f''(x^*) = \dots = f^{(n-1)}(x^*) = 0,$$

but $f^{(n)}(x^*) \neq 0$. Then $f(x^*)$ is

(i) a minimum value of $f(x)$ if $f^{(n)}(x^*) > 0$ and n is even; (ii) a maximum value of $f(x)$ if $f^{(n)}(x^*) < 0$ and n is even; (iii) neither a maximum nor a minimum if n is odd so in this case the point x^* is called a point of inflection. This is a required sufficiency condition for determining either the function's minimum or maximum position.

According to theorems 2.1 and 2.2, the required conditions for necessity and sufficiency must be achieved in order to determine the optimum locations. In other words, the basic steps to find the optimum point for any function of a single variable are to simply differentiate it once and set it to zero to be solved for x to determine x^* . Then the second derivative (n is even) is obtained to evaluate whether it is less than zero (concave), then a local maximum x^* exists, or greater than zero (convex), then a local minimum x^* exists, as illustrated in figure 2.1.

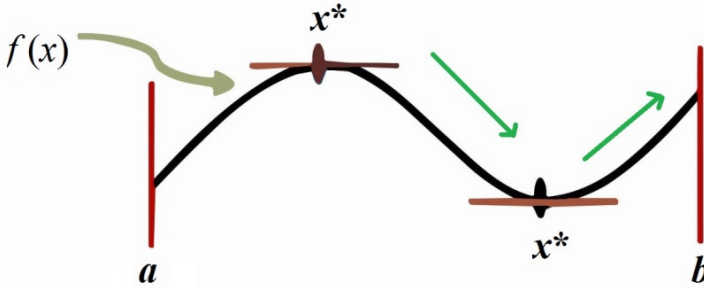


Figure 2.1

Example 2.1. Determine the local optimum points of the following function.

$$f(x) = 2x^3 - 3x^2 - 12x + 4$$

Solution:

Since $f'(x) = 6x^2 - 6x - 12 = 6(x + 1)(x - 2),$

$$f'(x) = 0 \text{ at } x = -1, x = 2.$$

The second derivative is

$$f''(x) = 12x - 6 = 6(2x - 1).$$

Since $f''(-1) = -18$, thus $x = -1$ is a local maximum. Therefore

$$f_{\max}(x) = f(x = -1) = 11.$$

Since $f''(2) = 18$, and thus $x = 2$ is a local minimum. Therefore

$$f_{\min}(x) = f(x = 2) = -16$$

and the maximum and minimum points are shown in figure 2.2.

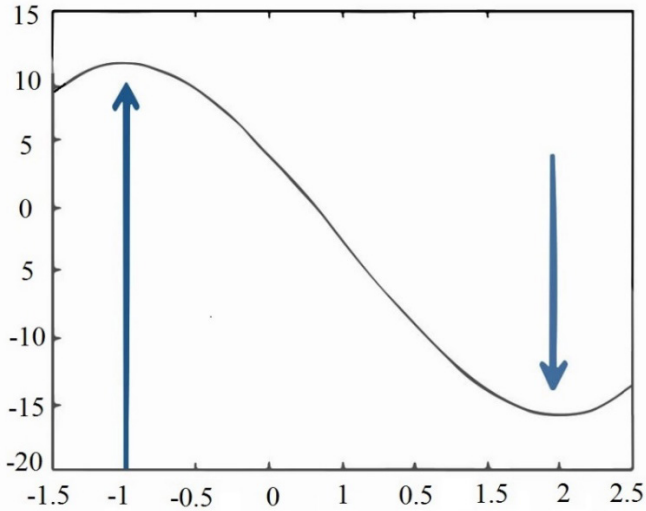


Figure 2.2

Finding the optimum points of any function with multivariables follows the same procedure; the only difference is that we are working with matrices. The required criteria for necessity and sufficiency conditions for a multivariable function's relative minimum (or maximum) points are provided by the next two theorems (Rao, 2019):

Theorem 2.3. If a function of multivariable $f(\mathbf{X})$ has a local minimum (or maximum) at $\mathbf{X} = \mathbf{X}^*$, and the $f'(\mathbf{X})$ exist at $\mathbf{X}^*$, then

$$\frac{\partial f}{\partial x_1}(\mathbf{X}^*) = \frac{\partial f}{\partial x_2}(\mathbf{X}^*) = \dots = \frac{\partial f}{\partial x_n}(\mathbf{X}^*) = 0,$$

it's necessary, which is still not adequately sufficient, in order to determine the function's optimum positions.

Theorem 2.4. If a function of multivariable $f(\mathbf{X})$ has a relative optimum at $\mathbf{X} = \mathbf{X}^*$, then the second partial derivative matrix (the Hessian matrix) of $f(\mathbf{X})$ evaluated at $\mathbf{X}^*$ is (i) positive definite when $\mathbf{X}^*$ is a relative minimum point, and (ii) negative definite when $\mathbf{X}^*$ is a relative maximum point. This is a sufficient condition for determining the optimum positions of the function.

For calculation simplicity, say we have a function of two different variables $f(x_1, x_2)$, then the Hessian matrix of $f(\mathbf{X})$ is presented by

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}.$$

Note: It is easy to evaluate the positive or negative definiteness of a matrix A of n -order using its determinants as

$$A_1 = |a_{11}|, A_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \dots, A_n = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$$

Only when every value in the matrices $A_1, A_2, A_3, \dots, A_n$ is positive, then the A matrix is considered positive-definite. On the other hand, if and only if the sign of A_j is $(-1)^j$ for $j = 1, 2, \dots, n$ then the matrix A is considered negative definite.

Example 2.2. Find the optimum locations of the following function

$$f(x_1, x_2) = 20x_1 + 26x_2 + 4x_1x_2 - 4x_1^2 - 3x_2^2$$

Solution:

According to the required necessity conditions for the $f(x_1, x_2)$ that's yield

$$\frac{\partial f}{\partial x_1} = 20 + 4x_2 - 8x_1 = 0, \quad \rightarrow x_1 = \frac{20 + 4x_2}{8},$$

$$\frac{\partial f}{\partial x_2} = 26 + 4x_1 - 6x_2 = 0, \quad \rightarrow x_1 = \frac{6x_2 - 26}{4}$$

After simplification

$$\frac{20 + 4x_2}{8} = \frac{6x_2 - 26}{4} \rightarrow x_2 = 9, \\ x_1 = 7.$$

The extreme points of the $f(x_1, x_2)$ are $x_1^* = 7, x_2^* = 9$. To determine if the solution is a minimum or a maximum, the sufficient condition is applied to $f(x_1, x_2)$. The partial derivatives of second order for $f(x_1, x_2)$ at (x_1^*, x_2^*) are presented by

$$\frac{\partial^2 f}{\partial x_1^2} = -8, \quad \frac{\partial^2 f}{\partial x_2^2} = -6, \quad \frac{\partial^2 f}{\partial x_1 x_2} = 4, \quad \frac{\partial^2 f}{\partial x_2 x_1} = 4$$

The H matrix of $f(x_1, x_2)$ is given by

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 x_2} \\ \frac{\partial^2 f}{\partial x_2 x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} -8 & 4 \\ 4 & -6 \end{bmatrix}$$

The determinants of H_1 and H_2 are

$$H_1 = |-8| = -8, \quad H_2 = \begin{vmatrix} -8 & 4 \\ 4 & -6 \end{vmatrix} = +32.$$

Since

$$\frac{\partial^2 f}{\partial x_1^2} = -8 < 0 \text{ and } \left(\frac{\partial^2 f}{\partial x_1^2} \frac{\partial^2 f}{\partial x_2^2} \right) - \left(\frac{\partial^2 f}{\partial x_1 \partial x_2} \frac{\partial^2 f}{\partial x_2 \partial x_1} \right) = 32 > 0,$$

that corresponds with the negative definite rule for the H matrix. Thus, $f(x_1, x_2)$ is negative-definite where (x_1^*, x_2^*) is located. Therefore

$$f_{max}(x_1, x_2) = 187$$

and the optimum points are shown in figure 2.3.

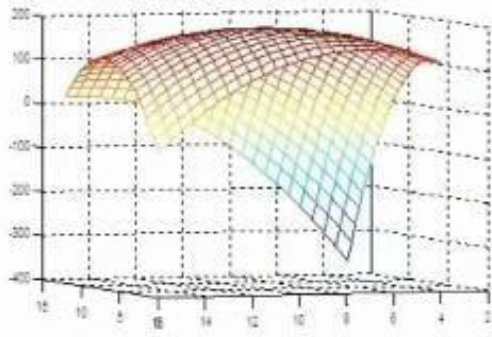


Figure 2.3

When the first derivatives $\partial f / \partial x_1 = \partial f / \partial x_2 = 0$, the H matrix may not be positive or negative definite at that point (x_1^*, x_2^*) . When that occurs, the point (x_1^*, x_2^*) is referred to as a saddle point since it isn't considered either the local maximum or the local minimum in this case. For example the following function $f(x_1, x_2) = x_1^2 - x_2^2$ the first derivatives are

$$\frac{\partial f}{\partial x_1} = 2x_1 = 0 \rightarrow x_1 = 0, \quad \frac{\partial f}{\partial x_2} = -2x_2 = 0, \rightarrow x_2 = 0$$

The partial derivatives of second-order of $f(x_1, x_2)$ at (x_1^*, x_2^*) are presented by

$$\frac{\partial^2 f}{\partial x_1^2} = 2, \quad \frac{\partial^2 f}{\partial x_2^2} = -2, \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = 0, \quad \frac{\partial^2 f}{\partial x_2 \partial x_1} = 0$$

The H of $f(x_1, x_2)$ is given by

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 x_2} \\ \frac{\partial^2 f}{\partial x_2 x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

The determinants of H_1 and H_2 are

$$H_1 = |2| = +2, \quad H_2 = \begin{vmatrix} 2 & 0 \\ 0 & -2 \end{vmatrix} = -4$$

Since

$$\frac{\partial^2 f}{\partial x_1^2} = 2 > 0, \text{ and } \left(\frac{\partial^2 f}{\partial x_1^2} \cdot \frac{\partial^2 f}{\partial x_2^2} \right) - \left(\frac{\partial^2 f}{\partial x_1 x_2} \cdot \frac{\partial^2 f}{\partial x_2 x_1} \right) = -4 < 0$$

Given that this matrix lacks positive and negative definiteness, the point (x_1^*, x_2^*) is a saddle point. It is marked in red on the figure 2.4 shown below

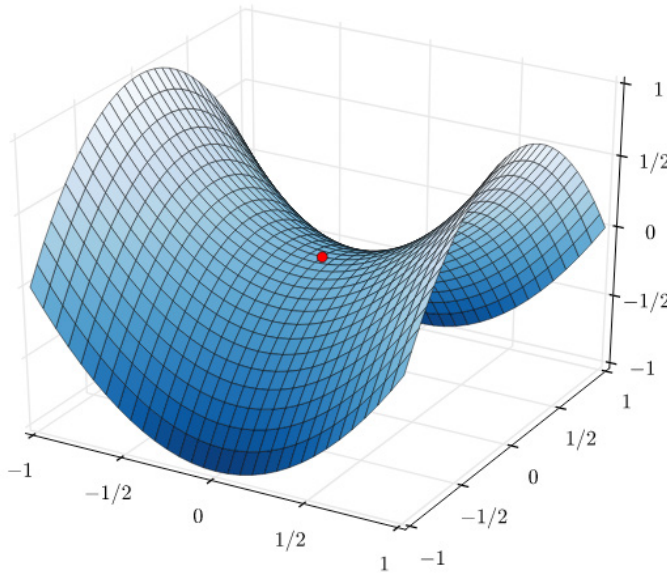


Figure 2.4

3. Gradient-Based Numerical Optimization Techniques

Gradient-based techniques lead the search by utilizing the gradient's information, or the objective function's direction. Until they reach a local optimum or a convergence requirement is satisfied, they begin at an initial random point and update iteratively by taking little steps along the gradient.

The distinction between all numerical techniques is the way they compute the step size along a specified search direction. The major advantages of gradient-based techniques can be stated as follows: they are quick and efficient when the objective function is smooth, convex, and differentiable. Additionally, they could enhance the search by exploiting the structure and features of the objective function. The primary drawback of gradient-based techniques is their sensitivity to the function's noise, step size, and initial point selection. Furthermore, the possibility of getting trapped at saddle points or local optima may prevent them from exploring the global optimum (Rao, 2019).

In this section, the two main gradient-based methods for solving single and multivariable optimization problems—steepest descent and Newton's method—are covered with examples.

3.1. Steepest Descent Technique

Particularly, the earliest technique for a nonlinear function's minimization is the classical steepest descent method. The gradient descent is also another name for this method. It is recognized as the first method for solving a nonlinear problem employing the gradient, as suggested by Cauchy in 1847 (Meza, 2010). As with all classical numerical techniques, the method begins with an initial point and continues by selecting the search directional path that creates a sharp angle with the gradient, which is considered to be negative, with specific steps to generate new estimates. However, as mentioned before, measuring the length of each step along a specified search directional path is the basis of nearly all of the minimization techniques. So the only challenge remaining is determining how long the step should be (Theodoridis, 2020). The steepest descent technique is considered efficient and simple, which is the main advantage of this method. The steps of the gradient descent technique can be summarized as follows:

1. Start with an initial point X_1 and set the iteration number as $i = 1$.
2. Calculate the search direction S_i as $S_i = -\nabla f_i = -\nabla f(X_i)$
3. Determine the step length λ_i^* in the direction S_i as

$$f(X_i + \lambda_i^* S_i), \quad \frac{df}{d\lambda} = 0 \rightarrow \lambda_i = \lambda_i^*$$

3. Find the new point as $X_{i+1} = X_i + \lambda_i^* S_i = X_i - \lambda_i^* \nabla f_i$

4. Test the new point, X_{i+1} , for optimality. If X_{i+1} is optimum, stop the process. Otherwise, set the new iteration number $i = i + 1$ and go back to step 2 (Rao, 2019).

Example 3.1. Apply minimization for

$$f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$$

by Steepest descent technique using the initial point $X_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Stop after the first three iterations, unless the convergence is satisfied before.

Solution:

Iteration 1:

$X_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, The gradient of $f(x_1, x_2)$ is :

$$\nabla f = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \rightarrow \nabla f_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Therefore,

$$S_1 = -\nabla f_1 = -\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

To determine X_2 , we must find the length of the first step, λ_1^* . For this, we minimize $f(X_1 + \lambda_1^* S_1)$ based on λ_1^* .

$$\begin{aligned} f(X_1 + \lambda_1^* S_1) &= f\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix} + \lambda_1^* \begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} -\lambda_1^* \\ \lambda_1^* \end{bmatrix}\right) \\ &= f(-2\lambda_1^* + \lambda_1^{*2}) \end{aligned}$$

we get

$$\frac{df}{d\lambda_1^*} = 0 \rightarrow -2 + 2\lambda_1^* = 0 \rightarrow \lambda_1^* = 1.$$

$$X_2 = X_1 + \lambda_1^* S_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} \rightarrow X_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Test X_2

$$\nabla f_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Not optimum, next iteration.

Iteration 2:

$$X_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, S_2 = -\nabla f_2 = -\begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

To minimize

$$\begin{aligned} f(X_2 + \lambda_2^* S_2) &= f\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix} + \lambda_2^* \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} -1 + \lambda_2^* \\ 1 + \lambda_2^* \end{bmatrix}\right), \\ &= f(-1 - 2\lambda_2^* + 5\lambda_2^{*2}), \end{aligned}$$

we get

$$\frac{df}{d\lambda_2^*} = 0 \rightarrow -2 + 10\lambda_2^* = 0 \rightarrow \lambda_2^* = 0.2$$

$$X_3 = X_2 + \lambda_2^* S_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} + 0.2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow X_3 = \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix}$$

Test X_3

$$\nabla f_3 = \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Not optimum, continue to the next iteration since the gradient's components are not zero.

Iteration 3:

$$X_3 = \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix}, \quad S_2 = -\nabla f_3 = -\begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix} = \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix}$$

To minimize

$$\begin{aligned} f(X_3 + \lambda_3^* S_3) &= f\left(\begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix} + \lambda_3^* \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix}\right) = f\left(\begin{bmatrix} -0.8 + 0.2\lambda_3^* \\ 1.2 + 0.2\lambda_3^* \end{bmatrix}\right), \\ &= f(-1.2 - 0.08\lambda_3^* + 0.04\lambda_3^{*2}), \end{aligned}$$

we get

$$X_4 = X_3 + \lambda_3^* S_3 = \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix} + 1 \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix} \rightarrow X_4 = \begin{bmatrix} -1 \\ 1.4 \end{bmatrix}$$

Test X_4

$$\nabla f(X_3) = \begin{bmatrix} -0.2 \\ -0.2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \nabla f_4$$

Not optimum, next iteration. Since $f_4 \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is not the optimum solution, we must go on to the next iteration in order to find the optimum one, but we stop here in accordance with the three iterations stated in the question. Unfortunately, this method is not primarily used since, in most real-world applications, it is known to be extremely slow and is rarely used in practice, regardless of its simplicity. Alternative, more efficient techniques like the Newton, conjugate gradient, and quasi-Newton methods are frequently utilized (Meza, 2010).

3.2. Newton's Method

A common classical optimization technique that uses gradients is Newton's method. Isaac Newton introduced it for solving nonlinear equations, and Joseph Raphson improved it later in 1690. That's why this technique is also

referred to as the Newton-Raphson method in the literature (Ypma, 1995). This method approximates the roots of the objective function by starting with an initial guess, then using tangent lines to create a set of estimates until reaching the optimum solution or the convergence criteria is satisfied (Polyak, 2007). Unlike the steepest decent method, the Newton method in its classic form has a fixed step size of 1, to obtain the optimum for any positive definite quadratic function, and the directional path of the search is provided by the Hessian matrix (Venter, 2010). This method's primary benefit is that it locates roots quickly and effectively. The steps of Newton's method can be stated down as follows:

1. Start with an arbitrary initial point $\mathbf{X}_1$ and set the iteration number as $i=1$.
2. Calculate the gradient as ∇f_i
3. Calculate the inverse of the H matrix H_i^{-1}
4. Calculate the search direction as $\mathbf{S}_i = -H_i^{-1}\nabla f_i$
5. Find the new point as $\mathbf{X}_{i+1} = \mathbf{X}_i - H_i^{-1}\nabla f_i = \mathbf{X}_i + \mathbf{S}_i$
6. Test the new point, $\mathbf{X}_{i+1}$, for optimality. If $\mathbf{X}_{i+1}$ is optimum, stop the process. Otherwise, set the new iteration number $i = i + 1$ and go back to step 2 (Rao, 2019).

Example 3.2. Minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$ by using Newton's method with the initial point $\mathbf{X}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Stop after the first three iterations, unless the convergence is satisfied before.

Solution:

Iteration 1:

The gradient of $f(x_1, x_2)$ is

$$\nabla f_1 = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

The partial derivatives of second-order of $f(x_1, x_2)$ at (x_1^*, x_2^*) are presented by

$$\frac{\partial^2 f}{\partial x_1^2} = 4, \quad \frac{\partial^2 f}{\partial x_2^2} = 2, \quad \frac{\partial^2 f}{\partial x_1 x_2} = 2, \quad \frac{\partial^2 f}{\partial x_2 x_1} = 2.$$

The H of $f(x_1, x_2)$ is

$$H_1 = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 x_2} \\ \frac{\partial^2 f}{\partial x_2 x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}.$$

$$H_1^{-1} = \frac{1}{(4 \times 2) - (2 \times 2)} \begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$

The directional path of the search S_1 is

$$S_1 = -H_1^{-1} \nabla f_1 \rightarrow S_1 = - \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1.5 \end{bmatrix}.$$

Determining X_2

$$X_2 = X_1 - H_1^{-1} \nabla f_1 = X_1 + S_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ -1.5 \end{bmatrix} = \begin{bmatrix} -1 \\ 1.5 \end{bmatrix} \rightarrow X_2 = \begin{bmatrix} -1 \\ 1.5 \end{bmatrix}$$

Test X_2

$$\nabla f(X_2) = \nabla f_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

It is optimum stop.

As a result, for this quadratic function, in a single iteration, the method successfully converged. In spite of this method's main advantages, it is not particularly helpful in real-world applications because of the difficulty of computing the elements of the Hessian matrix at every step. However, the majority of gradient-based techniques just use first-order gradient information.

4. Elimination Techniques

As previously indicated, if the problems contain non-continuous and perhaps non-differentiable functions, then classical techniques for optimization have restricted usage for several of the difficulties that arise in many scientific areas. For that reason, other numerical techniques, including elimination methods, can be helpful in this situation. In the elimination methods, the first interval of uncertainty, which is the interval at which the optimum solution must be included, is step-by-step decreased to the desired precision. The primary limitation of these techniques is their requirement for a unimodal objective function to be optimized. In order to optimize an objective function that is known to be multimodal, its range must be separated into multiple parts, each of which must be handled independently as a unimodal function. This allows for the use of elimination techniques. The elimination method procedure is the opposite

of classical techniques, in which the objective function's values are initially obtained at a various of the design variables, and conclusions regarding the optimum solution are then made. (Gill, et al, 1981). In this section, the two main popular elimination methods for solving unimodel optimization problems—Fibonacci and golden section—are covered with examples.

4.1. Fibonacci Elimination Method

Among all the elimination methods, the most well-known, efficient, and complex method, according to the literature, is the Fibonacci Method, which uses the famous Fibonacci sequence in the picking of pieces from the objective function to get optimized. Leonardo of Pisa, also known as Fibonacci, initially presented the Fibonacci numbers in 1202 (Scott & Marketos, 2014). The Fibonacci numbers, $\{F_n\}$, are defined as

$$F_0 = F_1 = 1, \quad F_n = F_{n-1} + F_{n-2}, n = 2, 3, 4, \dots$$

which generates the sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

The following requirements must be achieved for the Fibonacci search method to be applied to a real-world problem (Rao, 2019):

- It is necessary to know the first interval of uncertainty that contains the optimum.
- In the first uncertainty interval, the objective function that is being optimized must be unimodal.
- This technique is unable to determine the precise optimum. There will only be one interval that is known as a last interval of uncertainty.
- The number of objective function assesments to be used in the search or the necessary trials n needs to be determined in advance.

The steps of Fibonacci elimination method can be summarized as follows:

1. Define $L_0 = b - a$ the first interval of uncertainty in $a \leq x \leq b$ and n be the entire amount of trials that need to be carried out.
2. Define $L_2^* = \frac{F_{n-2}}{F_n} L_0$ and place the first two trials at points x_1 and x_2 . This gives

$$x_1 = a + L_2^*, \quad x_2 = b - L_2^*$$

3. Find $f(x_1)$ and $f(x_2)$ as shown in figure 4.1

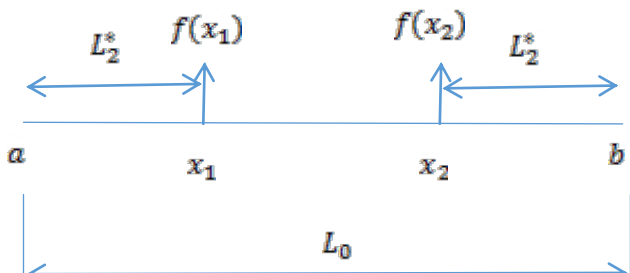


Figure 4.1

4. Compare $f(x_1)$ and $f(x_2)$

- If $f(x_1) < f(x_2)$, as shown in figure 4.2, delete $[x_2, b]$ and calculate $x_j = a + (x_2 - x_1)$, $j = 3, 4, \dots, n$

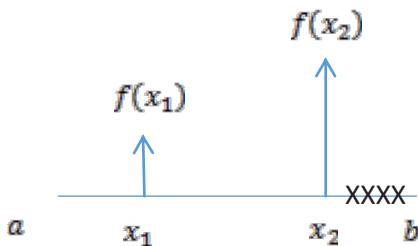


Figure 4.2

- If $f(x_1) > f(x_2)$, as shown in figure 4.3, delete $[a, x_1]$ and calculate $x_j = x_1 + (b - x_2)$, $j = 3, 4, \dots, n$

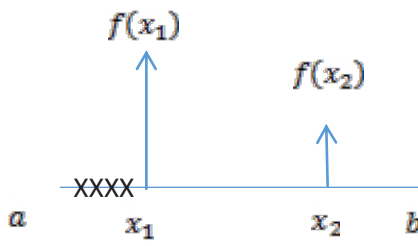


Figure 4.3

- If $f(x_1) = f(x_2)$ as shown in figure 4.4, calculate $x_j = a + \frac{F_{n-j}}{F_n} (b - a)$, $j = 3, 4, \dots, n$

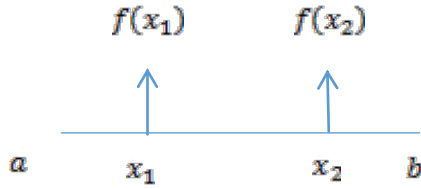


Figure 4.4

5. Continue the process by repeating the steps until the number of trials n is finished and the last interval $[x, x + L_n]$ is located.

6. Calculate the proportion of the last to the first interval of certainty $\frac{L_n}{L_0}$.

Example 4.1. Minimize $f(x) = e^x - \sin x$ in the interval $[-1, 1]$ by using the Fibonacci method with $n = 4$.

Solution:

Hence $n = 4$, $a = -1$, $b = 1$ and $L_0 = 1 - (-1) = 2$, that yield

$$L_2^* = \frac{F_{n-2}}{F_n} L_0 = \frac{2}{5} \times 2 = 0.8.$$

Thus, the positions of the first two trials are given by

$$x_1 = a + L_2^* = -1 + 0.8 = -0.2 \text{ with } f(x_1) = 1.0174$$

and

$$x_2 = b - L_2^* = 1 - 0.8 = 0.2 \text{ with } f(x_2) = 1.0227,$$

as shown in figure 4.5.

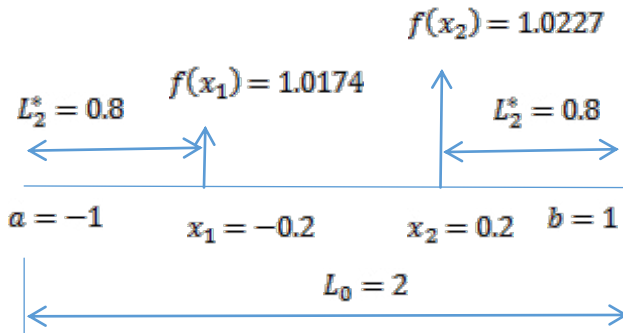


Figure 4.5

Since $f(x_1) < f(x_2)$, the interval $[x_2, b]$ will be deleted as shown in figure 4.6.

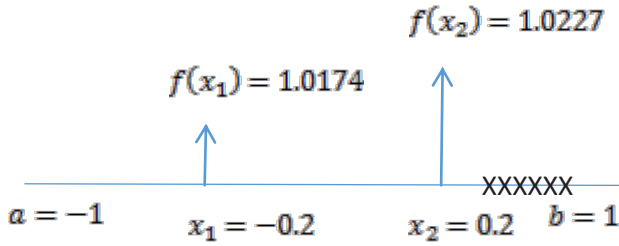


Figure 4.6

The uncertainty interval L_2 is calculated as $L_2 = 0.2 - (-1) = 1.2$ for the third trial which is placed at x_3 . The value of x_3 is given as

$$x_3 = a + (x_2 - x_1) = -1 + (0.2 - (-0.2)) = -0.6$$

with $f(x_3) = 1.1135$. Since $f(x_1) < f(x_3)$, the interval $[a, x_3]$ will be deleted as shown in figure 4.7.

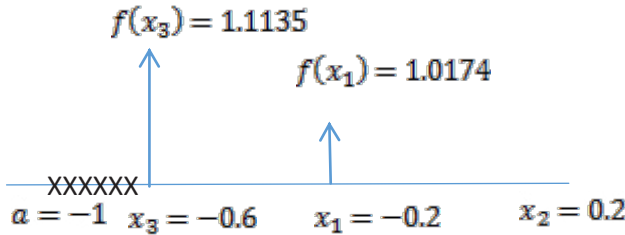


Figure 4.7

For the next trial the interval L_3 is calculated as $L_3 = 0.2 - (-0.6) = 0.8$ and it is located at x_4 . The value of x_4 is given as

$$x_4 = x_3 + (x_2 - x_1) = -0.6 + (0.2 - (-0.2)) = -0.2$$

Since $x_4 = x_1$, the x_4 is found again by the following formula

$$\begin{aligned} x_4 &= a + \frac{F_{n-j}}{F_n}(b - a) = x_3 + \frac{F_{4-4}}{F_4}(x_2 - x_3) \\ &= -0.6 + \frac{1}{5}(0.2 - (-0.6)) = -0.6 + 0.16 = -0.44 \end{aligned}$$

with $f(x_4) = 1.070$. Since $f(x_1) < f(x_4)$ the interval $[x_3, x_4]$ will be deleted as shown in figure 4.8.

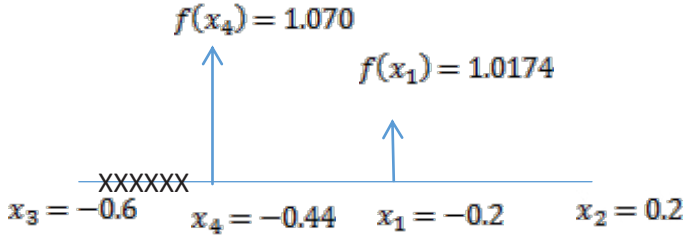


Figure 4.8

The last interval of uncertainty is obtained as

$$L_4 = [-0.44, 0.2] = 0.2 - (-0.44) = 0.64$$

as shown in the figure 4.9

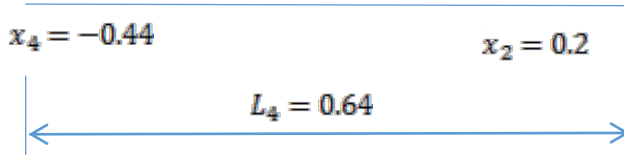


Figure 4.9

The proportion of the last to the first interval of uncertainty is

$$\frac{L_4}{L_0} = \frac{0.64}{2} = 0.32$$

as shown in figure 4.10.

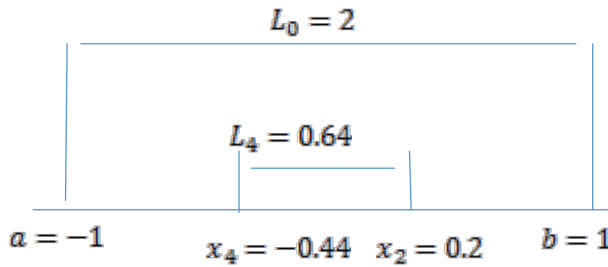


Figure 4.10

4.2. Golden Section Methods

The Golden Section Optimization Method is a unique optimization algorithm inspired by the golden ratio to find the optimum points for unimodal

objective functions by successfully narrowing the interval of certainty. This method, which is based on the Fibonacci sequence, is also known as the golden ratio method or the golden mean method. It makes use of the golden ratio, which is simply that if the length of a line is split into two uneven parts (a and b), the proportion of the whole ($a + b$) to the bigger part (a) is equal to the proportion of the bigger (a) to the lesser (b). Its expressed in symbols as

$$\gamma = \frac{a+b}{a} = \frac{a}{b} \cong 1.618$$

The same as Fibonacci elimination technique this method based on the assessment of the objective function at different points in the certainty interval and reducing it as much as possible to reach the optimum.

The steps of golden section elimination method can be summarized like this:

1. Define $L_0 = b - a$ the first interval of uncertainty in $a \leq x \leq b$
2. Define

$$L_2^* = \frac{L_0}{\gamma^2} = \frac{L_0}{1.618^2} = 0.382L_0$$

and position the first two trials at points x_1 and x_2 , which provides

$$x_1 = a + L_2^*, \quad x_2 = b - L_2^*$$

3. Find $f(x_1)$ and $f(x_2)$, as shown in figure 4.11.

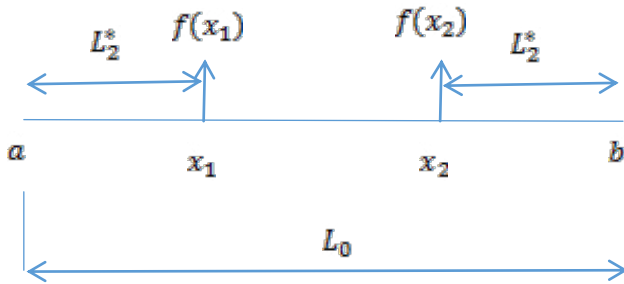


Figure 4.11

4. Compare $f(x_1)$ and $f(x_2)$

- If $f(x_1) < f(x_2)$, as shown in figure 4.12, delete $[x_2, b]$ and calculate

$$x_j = a + (x_2 - x_1), j = 3, 4, \dots, n.$$

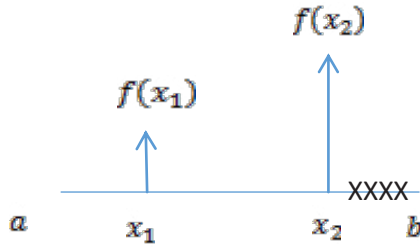


Figure 4.12

- If $f(x_1) > f(x_2)$, as shown in figure 4.13, delete $[a, x_1]$ and calculate $x_j = x_1 + (b - x_2)$, $j = 3, 4, \dots, n$.

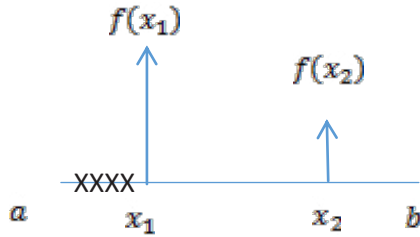


Figure 4.13

- If $f(x_1) = f(x_2)$, as shown in figure 4.14, calculate $x_j = a + \frac{(b-a)}{\gamma^j}$, $j = 3, 4, \dots, n$.

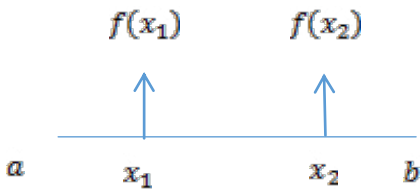


Figure 4.14

5. Continue the process by repeating the steps until the number of trials n is finished and the last interval $[x, x + L_n]$ is located.

6. Calculate the proportion of the last to the first interval of certainty $\frac{L_n}{L_0}$.

Example 4.2. Minimize $f(x) = e^x - \sin x$ in the interval $[-1,1]$ by using the golden section method with $n = 4$.

Solution:

Hence $n = 4$, $a = -1$, $b = 1$ and $L_0 = 1 - (-1) = 2$, that yield

$$L_2^* = 0.382 \times L_0 = 0.382 \times 2 = 0.764$$

Consequently, the first two trials' positions are provided by

$$x_1 = a + L_2^* = -1 + 0.764 = -0.236 \text{ with } f(x_1) = 1.0236$$

and

$$x_2 = b - L_2^* = 1 - 0.764 = 0.236 \text{ with } f(x_2) = 1.0324,$$

as shown in figure 4.15.

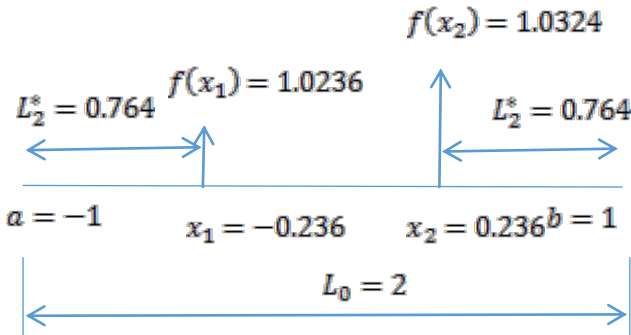


Figure 4.15

Since $f(x_1) < f(x_2)$, the interval $[x_2, b]$ will be deleted as shown in figure 4.16

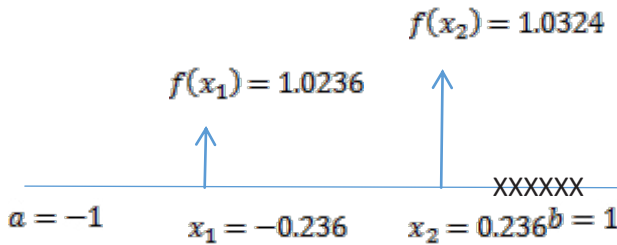


Figure 4.16

The uncertainty interval L_2 is calculated as $L_2 = 0.236 - (-1) = 1.236$ for the third trial which is placed at x_3 . The value of x_3 is given as

$$x_3 = a + (x_2 - x_1) = -1 + (0.236 - (-0.236)) = -0.528$$

with $f(x_3) = 1.0936$. Since $f(x_3) < f(x_1)$, the interval $[a, x_3]$ will be deleted as shown in figure 4.17.

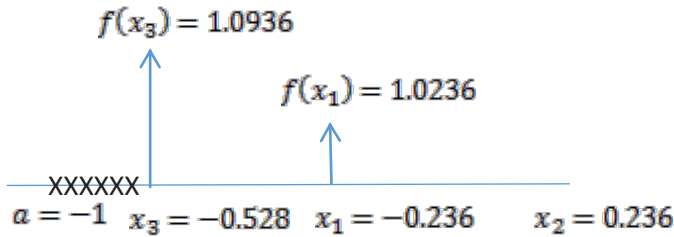


Figure 4.17

For the next trial the interval L_3 is calculated as

$$0.236 - (-0.528) = 0.764$$

and it is located at x_4 . The value of x_4 is given as

$$x_4 = x_3 + (x_2 - x_1) = -0.528 + (0.236 - (-0.236)) = -0.056$$

with $f(x_4) = 1.0015$. Since $f(x_4) < f(x_1)$ the interval $[x_3, x_1]$ will be deleted as shown in figure 4.18

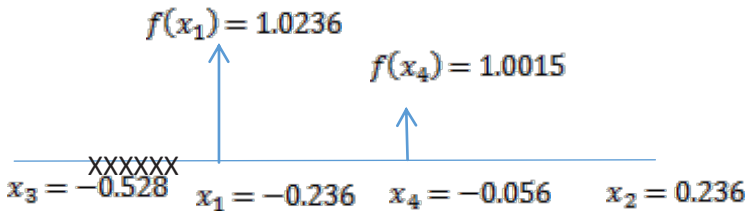


Figure 4.18

The last interval of uncertainty is obtained as

$$L_4 = [-0.236, 0.236] = 0.236 - (-0.236) = 0.472,$$

as shown in the figure 4.19.

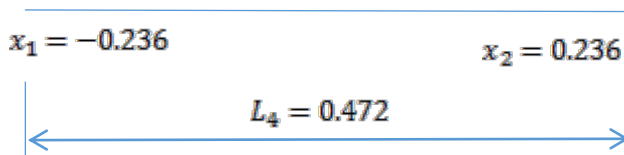


Figure 4.19

the proportion of the last to the first interval of uncertainty is

$$\frac{L_4}{L_0} = \frac{0.472}{2} = 0.236,$$

as shown in figure 4.20.

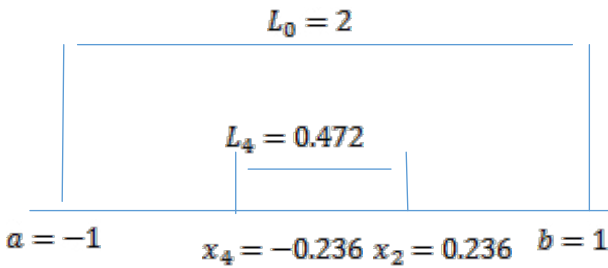


Figure 4.20

5. Conclusion

The classical optimization techniques are analytical techniques that use differential calculus to find the optimum of continuous and differentiable functions. According to the optimization theory, the required conditions for necessity and sufficiency must be achieved in order to detect a function's optimum positions when applying classical optimization techniques to a single or multivariable function. Classical optimization techniques based on gradient methods are used to find optimal solutions for nonlinear objective functions. All these techniques rely on the directional path of the search and the length of each step when optimizing functions, and calculating them is the main distinction between all gradient methods. One of the first techniques for minimizing an ordinary nonlinear function is the steepest descent method. It is a first-order method, meaning that at each step, it only takes into consideration the objective function's first derivative. However, in spite of its simplicity, it is known to be quite slow to converge, and it is not often used in practice. Alternatively, more fast and efficient techniques like Newton's method are used. Nevertheless, Newton's method is a two-order method, as it uses each of the first and second derivatives, which makes it costly for real-world applications due to the complexity of computing the elements of the Hessian matrix at every step. Elimination techniques can also be used to optimize nonlinear unimodal functions, regardless of whether they are discontinuous or non-differentiable. Between the elimination methods, Fibonacci and the golden section techniques are fast, accurate, robust, and derivative-free. However, unlike classical

techniques, elimination methods are not applied to multimodal functions. All classical optimization methods are sensitive to the selected initial point, which means that in some cases, other methods based on random search, like meta-heuristic algorithms, can be more efficient and can be recommended to optimize nonlinear objective functions.

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